Genetic Algorithm Guide

Inputs

Live MEC resources from the digital twin (LOCAL, MEC-XXX*, CLOUD)
with cpu_usage, ram_usage, latency, energy besides per-device
security_risk / privacy_loss scores.
Only ONLINE, non-revoked MEC devices are eligible (OFFLINE / REVOKED
are never assigned tasks; their tasks are requeued automatically).

Optimizer

Scheduler.proposed_ga_schedule() - steady-state GA, population 40,
generations 30, multi-objective fitness:

```
fitness = w1*latency + w2*energy + w3*security_risk + w4*privacy_loss
```

Parents selected by tournament; children via crossover + mutation;
elitism keeps the best chromosome. Returns assignments + metrics
(best_fitness, generation, convergence_time).

Dispatch

POST /api/tasks/dispatch/ {population, generations}
run_ga_dispatch() marks each task ASSIGNED + creates history entries
(PENDING/QUEUED -> ASSIGNED). No ONLINE MEC => tasks stay QUEUED,
they are never routed to offline devices.

Task lifecycle

PENDING -> QUEUED -> ASSIGNED -> RUNNING -> (COMPLETED / FAILED)
Agents poll /api/mec/<id>/next-task/ and report results via
/api/mec/<id>/report-result/. Offline/revoked devices are requeued.